Space Emanation Theory

Source, Capacity and the Luminosity History

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Abstract

SET treats a specified collection of baryonic matter as one source and describes its participation in a finite field capacity. This paper develops the luminosity history of a homogeneous cosmological realization. Its source-evolution principle states that a conserved material domain changes its physical volume at the rate Q in the same reference time used by capacity. Together with the central collective burden and a causal source front, that principle selects a definite clock and luminosity history for specified source inputs and a front initial state.

$$Q_D^2 = 24\pi G M_D V_D, \qquad \frac{dV_D}{dT} = Q_D, \qquad dt = \sqrt{1-u}\, dT \tag{1}$$

The derivation connects four readings of the same history: source-volume evolution, accumulated central burden, photon propagation and the common-energy constraint. It produces an explicit source-radius equation, its exact integral, a source-photon energy identity and a direct differential equation for luminosity distance. The retained energy constraint gives the corresponding relation (1−u)E_exp = E_b and specifies the field work needed to sustain it.

The source-radius calculation reproduces all fifteen archived Pantheon+ and DES-Dovekie brightness contrasts with a maximum numerical difference below 7.1×10^{-12}. Their inputs, predictions and graphs are preserved. The new result is the explicit selection within the stated cosmological realization and its connection to energy accounting. The source-evolution principle is stated as a physical premise; the separate compatibility test identifies what its realization in a local interacting field must supply.

The main results

1. A modern source-evolution principle with an explicit domain and clock.
2. A source-radius history that reproduces the existing luminosity calculation.
3. Direct source-photon and optical capacity equations.
4. An exact lapse comparison and the required source-energy exchange.

1. One source law, one capacity relation

SET begins with the participation of source burden, translational motion and local event progression in one field capacity. For a clock moving relative to the local field reading,

$$S^2 + V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2, \qquad d\tau = \beta\, dT \tag{2}$$

For the comoving clock used in this homogeneous calculation, the translational term vanishes. We write its progression as α and the central collective burden as S_c:

$$A \equiv \alpha^2 = 1 - u, \qquad u \equiv \frac{S_c^2}{c^2}, \qquad dt = \alpha\, dT \tag{3}$$

The source law for a specified domain Ω is

$$Q_\Omega^2 = 24\pi G M_\Omega V_\Omega, \qquad Q_\Omega = M_\Omega \sqrt{\frac{24\pi G}{\rho_\Omega}} \tag{4}$$

Here ρ_Ω = M_Ω/V_Ω. For a sphere, V = 4πR³/3, so this is the same law written as Q² = 32π²GMR³. Q has units of volume per time. Changing the description from a planet to a disjoint collection of sources does not change the algebraic law. The specified domain and the location at which its response is read must remain explicit.

Two domains with different bookkeeping

The subscript D denotes a material collection with fixed baryonic mass. Its volume can evolve. The subscript h denotes the advancing spherical source domain; it acquires additional matter as its boundary moves through the distribution. The same law applies to both.

$$\frac{Q_D^2}{V_D} = 24\pi G M_D, \qquad 2\frac{dQ}{Q} = \frac{dM}{M} + \frac{dV}{V} \tag{5}$$

An unchanged source law therefore does not require an unchanged numerical Q when its domain volume or enclosed mass changes. The fixed-mass source strength is Q_D²/V_D. This distinction is necessary when continuous sourcing is followed through a growing cosmological domain.

2. Why the central reading contains 3GM/R

Take a uniform spherical source domain of physical radius R_h and density ρ at the time of the reading. The retained isotropic shell response gives a central burden contribution 2G dm/r from a shell at radius r. Thus

$$dm = 4\pi\rho r^2\,dr, \qquad S_c^2 = 2G\int_0^{R_h}\frac{dm}{r} \tag{6}$$

$$S_c^2 = 8\pi G\rho\int_0^{R_h} r\,dr = 4\pi G\rho R_h^2 = \frac{3GM_h}{R_h} \tag{7}$$

The coefficient 3 comes from including the shells throughout the source domain and reading their combined response at its centre. It is not a coefficient adjusted to supernova brightness. This step uses the isotropic source-response kernel retained in the homogeneous application, in addition to the finite-domain source law.

$$M_h = \tfrac{4\pi}{3}\rho R_h^3, \qquad S_b^2 = \frac{2GM_h}{R_h}, \qquad S_c^2 = \tfrac{3}{2}S_b^2 \tag{8}$$

The surface reading S_b and central reading S_c concern the same source but different locations. For this sphere, Q_h/(4πR_h²) gives S_b. Substituting the central S_c in that same surface formula would mix two different readings.

The growing-radius capacity scale

If density is held fixed while the source radius is enlarged, the central response rises linearly with radius:

$$S_c(R) = R\sqrt{4\pi G\rho}, \qquad R_{\mathrm{cap}} = \frac{c}{\sqrt{4\pi G\rho}} \tag{9}$$

$$Q(R_{\mathrm{cap}}) = \sqrt{\frac{2c^6}{3G^2\rho^2}} \tag{10}$$

This is the algebraic capacity scale suggested by the collective-source picture. An evolving history must also evolve density. The next steps determine what that distinction does to the result. A local equality S_c = c does not by itself establish which photons reach an observer; that requires the photon trajectory.

3. The clock supplied by the source law

For the conserved material collection D, first construct a source-volume variable T_Q by integrating the volume increment divided by its source rate. This step is algebraic; the next section identifies its physical evolution clock. The positive expanding branch gives

$$dT_Q \equiv \frac{dV_D}{Q_D} = \frac{dV_D}{\sqrt{24\pi G M_D V_D}} \tag{11}$$

$$T_Q = \sqrt{\frac{V_D}{6\pi G M_D}} = \frac{1}{\sqrt{6\pi G\rho}}, \qquad \frac{dV_D}{dT_Q} = Q_D \tag{12}$$

The displayed origin is the formal zero-volume origin of this source-volume variable. Its differential is unchanged by a constant shift. At a given density, multiplying the representative domain’s mass and volume by the same factor leaves T_Q unchanged. The construction therefore does not depend on the arbitrary size of a homogeneous subdivision.

$$\rho = \frac{1}{6\pi G T_Q^2}, \qquad H_b \equiv \frac{Q_D}{3V_D} = \frac{2}{3T_Q} \tag{13}$$

Since V_D is proportional to a^3 for a conserved homogeneous collection, its scale factor obeys

$$\frac{a}{a_0} = \left(\frac{T_Q}{T_{Q0}}\right)^{2/3} \tag{14}$$

When the enclosed mass changes

The advancing domain h contains new matter. Differentiating the same density-defined variable while keeping this change gives

$$dT_Q = \frac{dV_h - (V_h/M_h)\,dM_h}{Q_h} \tag{15}$$

The second term removes volume growth caused solely by enclosing additional mass at the same density. This is how the source clock distinguishes the evolution of existing material from the advance of the source boundary. At a zero-radius launch, T_Q is evaluated from the finite representative-domain density, rather than from the front’s zero mass and zero volume.

4. The cosmological source-evolution principle

The source law fixes Q for a specified mass and volume. Its units alone do not say which physical volume changes. The homogeneous cosmological realization retains the following source-evolution principle: a conserved material collection changes its physical volume at the rate supplied by its source law, in the reference time that enters the capacity relation.

$$\frac{1}{V_D}\frac{dV_D}{dT} = \frac{Q_D}{V_D} = \sqrt{24\pi G\rho_D} \tag{16}$$

This is the modern form of SET's earlier description of sourcing as cosmological volume production. It fixes both the domain and the clock. D is a freely evolving homogeneous material domain, not the advancing front h and not the material interior of a bound planet. Appendix A separates production, temporal storage and boundary motion in the earlier formulation.

Comparing this principle with dT_Q = dV_D/Q_D gives dT = dT_Q. Choose the same origin, so T = T_Q. The comoving capacity clock and the source law then give

$$\frac{dt}{dT_Q} = \alpha = \sqrt{1-u}, \qquad H = \frac{1}{3V_D}\frac{dV_D}{dt} = \frac{Q_D}{3V_D\alpha} \tag{17}$$

The source-volume identity now has a physical evolution meaning. Within this realization, the source principle fixes the material pace relative to T; capacity fixes the material clock relative to T. Their combination fixes the pace measured by that clock.

One source increment, two clock readings

$$Q_{\mathrm{norm},\,D} = \frac{Q_D}{\alpha}, \qquad Q_D\,dT_Q = Q_{\mathrm{norm},\,D}\,dt \tag{18}$$

Q normalized is the same process expressed per second of the slower material clock. The conversion introduces no additional source. Photons are propagated through this clock history; the receiving photon does not need a proper clock for the surrounding material clocks and field to define the intervals.

The choice dV_D/dT = Q_D is a stated cosmological source principle. It is not derived from the scalar invariant alone. The following equations show what it selects when combined with the explicitly retained homogeneous response and front description.

5. A causal source front and its mass

Retain the homogeneous physical spatial reading $d\ell^2 = a^2 d\chi^2$ and the common limiting speed c relative to material clocks. An outward radial null front obeys

$$\frac{d\chi_h}{dt} = \frac{c}{a}, \qquad R_h = a\chi_h \tag{19}$$

Differentiating its physical radius gives the front equation used in the luminosity application:

$$\dot{R}_h = HR_h + c \tag{20}$$

A dot denotes d/dt. The term c is local front propagation; HR_h accounts for the changing physical spatial scale. This calculation identifies that null accumulation domain, launched from a specified initial event, with the source domain whose central burden sets local capacity. The central kernel is an equal-time uniform readout of that domain; a fully retarded inhomogeneous kernel is a separate field calculation.

Material conservation and boundary sweep

$$\dot{\rho} = -3H\rho, \qquad M_h = \frac{4\pi}{3}\rho R_h^3, \qquad V_h = \frac{4\pi}{3}R_h^3 \tag{21}$$

$$\dot{M}_h = 4\pi\rho R_h^2 c, \qquad \dot{V}_h = 3HV_h + 4\pi R_h^2 c \tag{22}$$

The first equality follows by differentiating M_h: dilution of the existing material cancels its contribution from the HR_h term, leaving the new mass swept up at the front. With $A_h = 4\pi R_h^2$, the source-volume form is

$$\frac{dV_h}{dT_Q} = Q_h + \alpha c A_h \tag{23}$$

The full moving volume therefore has both a material-evolution term and a swept-volume term. Applying $dV_h/dT_Q = Q_h$ to this entire advancing region would omit the latter. The law $dV_D/dT_Q = Q_D$ applies directly to the fixed material collection.

6. Accumulation, dilution and the radius equation

Now differentiate the central burden u = 4πGρR_h²/c². Material conservation and null-front growth give

$$\dot{u} = -Hu + \frac{8\pi G\rho R_h}{c} \tag{24}$$

$$\frac{\dot{u}}{u} = \frac{2c}{R_h} - H \qquad (R_h > 0) \tag{25}$$

This is the balance in the collective-source picture. The front includes new shells and raises the burden. Evolution of the material distribution dilutes it. The undivided equation remains regular at the zero-radius launch.

The source clock makes density explicit. Substitution in the central response gives

$$u = \frac{2R_h^2}{3c^2T_Q^2} \tag{26}$$

Multiplying the null-front equation by dt/dT_Q = √(1−u) and using H = 2/[3T_Q√(1−u)] now yields

$$\frac{dR_h}{dT_Q} = \frac{2R_h}{3T_Q} + c\sqrt{1 - \frac{2R_h^2}{3c^2T_Q^2}} \tag{27}$$

The source law, the central burden reading, the source-evolution principle and the null front have become one evolution equation. It determines the front’s history once its initial state is supplied. No luminosity coefficient has been introduced in reaching it.

A dimensionless form

$$X \equiv \frac{R_h}{cT_Q}, \qquad \frac{dX}{d\ln T_Q} = -\frac{X}{3} + \sqrt{1 - \frac{2X^2}{3}} \tag{28}$$

This form is convenient for integration and for seeing the response’s limiting behavior. Its physical capacity range is 0 ≤ X < √(3/2).

7. The exact luminosity-history equation

The burden variable is u = 2X²/3, while N ≡ ln a satisfies dN = (2/3)d ln T_Q. Therefore

$$\frac{du}{dN} = 2X\frac{dX}{d\ln T_Q} = -u + \sqrt{6u(1-u)} \tag{29}$$

This is precisely the burden history used in the archived luminosity calculation. It has now been reached by following the source radius and the density-defined source clock. For a regular integration variable on the lower branch, write

$$y \equiv \sqrt{\frac{u}{1-u}}, \qquad \frac{dy}{dN} = \frac{(1+y^2)(\sqrt{6}-y)}{2} \tag{30}$$

For k = √6, separation uses the elementary partial fractions

$$\frac{2}{(1+y^2)(k-y)} = \frac{2}{7}\left[\frac{1}{k-y} + \frac{y+k}{1+y^2}\right] \tag{31}$$

Integrate from the zero-response front y_i = 0 at a_i. The exact result for 0 ≤ y < √6 is

$$\ln\frac{a}{a_i} = \frac{2}{7}\left[-\ln\frac{\sqrt{6}-y}{\sqrt{6}} + \frac{1}{2}\ln(1+y^2) + \sqrt{6}\arctan y\right] \tag{32}$$

Differentiating this expression recovers the separated equation. Once the present burden is fixed by the source inputs, it determines a_i/a_0 for the zero-front branch. The launch ratio is not an additional fitted parameter.

Preserving the original front at launch

The u equation alone is not Lipschitz at u = 0 and admits mathematical waiting branches. The original radius equation fixes immediate front growth: at R_h = 0 and finite T_Q, dR_h/dT_Q = c. The regular y equation preserves this choice. The radius or y equation should therefore be used to initiate the calculation.

8. The source quantities along the history

Once u(T_Q) is known, mass, radius and Q follow algebraically. Start with the central response and the uniform density:

$$\rho = \frac{1}{6\pi G T_Q^2}, \qquad R_h = cT_Q\sqrt{\frac{3u}{2}} \tag{33}$$

$$M_h = \frac{4\pi}{3}\rho R_h^3 = \frac{2R_h^3}{9GT_Q^2} = \frac{c^3 T_Q}{\sqrt{6}G}u^{3/2} \tag{34}$$

$$Q_h = 12\pi G T_Q M_h, \qquad Q_{\mathrm{norm},\,h} = \frac{Q_h}{\sqrt{1-u}} \tag{35}$$

These expressions show how the same source law follows the changing collective domain. Q_h grows as new matter enters and its defining volume changes. No new source species appears in these identities.

The frozen reception benchmark

Quantity	Value
M_h0	1.5005279083 × 10⁵³ kg
R_h0	4.3992396698 × 10²⁶ m = 46.5 billion light-years
ρ_0	4.2074852219 × 10⁻²⁸ kg/m³
Q_h0	5.1893886387 × 10⁶² m³/s
u_0; α_0	0.7598922704; 0.4900078873
T_Q0	43.55432725 Gyr in source-reference units
H_b0; H_0	14.96663; 30.54365 km s⁻¹ Mpc⁻¹

These are the previously frozen, cosmology-conditioned source benchmark inputs and their derived quantities. They are reused here unchanged; they are not a new independent measurement of the absolute source scale. The grouped brightness comparison below cancels its common distance normalization.

Independent radius integration recovers the reception radius and mass to relative differences of 1.7 × 10⁻¹⁴ and 5.0 × 10⁻¹⁴. This checks the consistency of the source-radius reconstruction.

9. What the evolving collective source approaches

The dimensionless radius equation has one positive equilibrium. Solving its right-hand side equal to zero gives

$$X_* = \frac{3}{\sqrt{7}}, \qquad u_* = \frac{6}{7}, \qquad \alpha_* = \frac{1}{\sqrt{7}} \tag{36}$$

Below that equilibrium the burden grows; above it the burden decreases. The luminosity branch approaches S_c/c = √(6/7) ≈ 0.925820. Its clock remains real and positive. Fixed-density enlargement and evolution with changing density therefore give different limits under the same source law.

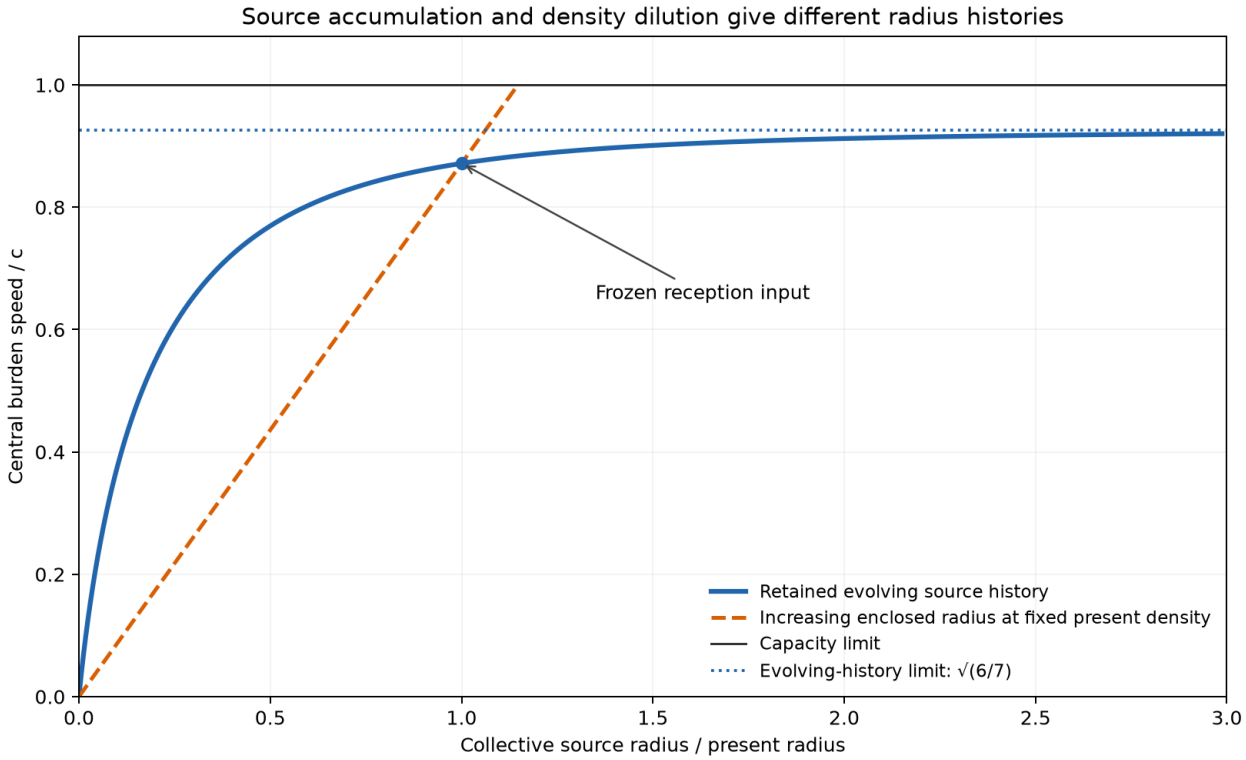


Figure 1. Model histories with the same reception benchmark. The fixed-density construction reaches the central capacity scale. The evolving source includes density dilution and approaches S_c/c = √(6/7). These curves are calculations, not observational data.

$$HR_h \longrightarrow 2c, \qquad \dot{R}_h \longrightarrow 3c, \qquad \rho \propto R_h^{-2} \tag{37}$$

The areal-radius rate 3c combines front propagation with the changing physical scale. The local propagation term remains c. At this limit M_h/R_h is constant. The evolution therefore regulates the central burden through the distribution’s density history, rather than requiring it to rise without limit.

10. From the clock history to photon brightness

Use the retained homogeneous photon coupling with comoving momentum p_χ. Its reference-time Hamiltonian and material-clock photon energy are

$$\mathcal{H}_\gamma = \alpha \frac{c|p_\chi|}{a}, \qquad E_\gamma = \frac{\mathcal{H}_\gamma}{\alpha} = \frac{c|p_\chi|}{a} \tag{38}$$

Spatial homogeneity conserves p_χ. Thus the photon energy ratio and the propagation interval give

$$\frac{E_o}{E_e} = \frac{a_e}{a_o} = \frac{1}{1+z}, \qquad \chi = c\int_{t_e}^{t_o} \frac{dt}{a} = c\int \frac{\alpha\, dT_Q}{a} \tag{39}$$

$$\frac{dt_o}{dt_e} = \frac{a_o}{a_e} = 1 + z \tag{40}$$

The arrival-interval relation follows by varying the endpoints of the same null path between comoving emitter and receiver. For conserved photons spreading over the flat homogeneous beam area 4π(a_oχ)², one factor of 1+z reduces each photon's energy and a second reduces the arrival rate:

$$F_o = \frac{L_e}{4\pi (a_o\chi)^2 (1+z)^2}, \qquad D_L = (1+z) a_o \chi \tag{41}$$

$$D_L(z) = (1+z)c\int_0^z \frac{dz'}{H(z')}, \qquad H(z) = H_{b0}\frac{(1+z)^{3/2}}{\alpha(z)} \tag{42}$$

The burden history enters through the clock-weighted distance integral. The source exposure identity Q dT_Q = Q_normalized dt is already respected in this calculation. An additional Q/α brightness multiplier after integrating this history would add a response that the photon derivation above has not supplied.

The physical content is specific: a collective source history changes clock progression, the history determines photon travel and arrival, and those quantities determine luminosity distance. Received brightness is obtained from that chain.

11. A direct source-photon energy identity

The photon calculation can be written entirely in terms of the source rate of the conserved material domain. Follow the same D through two epochs. Its baryonic mass M_D is fixed and its physical volume scales as a³. The source law therefore gives

$$Q_D^2 = 24\pi G M_D V_D, \qquad V_D \propto a^3, \qquad Q_D \propto a^{3/2} \tag{43}$$

The preceding photon Hamiltonian gives the locally measured photon energy E_γ ∝ a⁻¹. Combining the two relations eliminates a:

$$E_\gamma^3 Q_D^2 = \text{constant} \tag{44}$$

Between emission e and observation o, this becomes

$$\frac{E_o}{E_e} = \left(\frac{Q_{D,e}}{Q_{D,o}}\right)^{2/3}, \qquad 1 + z = \left(\frac{Q_{D,o}}{Q_{D,e}}\right)^{2/3} \tag{45}$$

The equation relates photon energy to the changing source reading of a fixed material collection. It must not be applied to Q_h without accounting for the mass swept into the advancing domain. It uses the unnormalized Q_D of the source law; its evolution and the clock conversion have already been specified.

What this identity contributes

This is a compact source-language expression for the redshift in the retained history. The result follows from the source law and the homogeneous photon coupling, with no separate brightness multiplier. It also distinguishes the photon's endpoint energy change from the distance it travels. The endpoint ratio alone does not determine the received flux: the travel integral, arrival intervals and beam area remain part of the calculation.

$$Q_{D,e}, Q_{D,o} \longrightarrow z, \qquad \alpha(z), H(z) \longrightarrow D_L(z) \tag{46}$$

The next reduction incorporates that travel history directly into a single optical equation. It makes the selected luminosity calculation accessible without first solving for the source radius as a function of time.

12. The optical equation in source and capacity variables

Use the homogeneous cosmological redshift, set a_o = 1 and write Z = 1+z. Define D = D_L/Z and h = H_b0/c, where H_b0 = √(8πGρ_0/3). Then D is the present physical radial distance and h has inverse-length units. The photon integral gives

$$D_z \equiv \frac{dD}{dz} = \frac{c}{H} = \frac{\alpha}{hZ^{3/2}} \tag{47}$$

The source front and the inward photon use the same conformal interval. Subtract the interval between emission and reception from the present front radius R_h0. Its emission-epoch radius and the density law are

$$R_h(z) = \frac{R_{h0} - D(z)}{Z}, \qquad \rho(z) = \rho_0 Z^3 \tag{48}$$

Insert these expressions into the uniform central burden u = 4πGρR_h²/c². Capacity then relates the optical derivative to the source front:

$$u = \tfrac{3}{2} h^2 Z (R_{h0} - D)^2 \tag{49}$$

$$h^2 \left[Z^3 D_z^2 + \tfrac{3}{2} Z (R_{h0} - D)^2 \right] = 1 \tag{50}$$

Choosing the forward distance branch gives the direct luminosity equation

$$D_z = \frac{\sqrt{1 - \tfrac{3}{2} h^2 Z (R_{h0} - D)^2}}{hZ^{3/2}}, \qquad D(0) = 0 \tag{51}$$

For supplied h and R_h0, its right-hand side is smooth while A > 0, giving a locally unique distance history. Apply it only while 0 ≤ D(z) ≤ R_h0 and A > 0, so emission lies on or after the retained front launch. These are the same reception inputs used by the radius calculation. No distance coefficient is fitted by rewriting the equation.

This equation uses one cosmological z to expose the algebra. In the catalogue calculation, the measured heliocentric redshift remains in the luminosity prefactor and the Hubble-diagram redshift remains in the integral, as detailed in the worked example.

13. What fixes this optical history

The role of the source principle can be checked without assuming its answer. For a general positive material-volume pace in the capacity reference time, define the diagnostic ratio λ. It records a possible history rather than introducing a new fitted parameter:

$$\lambda \equiv \frac{dV_D/dT}{Q_D}, \qquad H = \frac{\lambda H_b}{\alpha} \tag{52}$$

Keep the same density conservation, uniform central kernel, null front and photon coupling. Repeating the optical elimination gives

$$h^2\left[\lambda^2 Z^3 D_z^2 + \tfrac{3}{2} Z (R_{h0} - D)^2\right] = 1 \tag{53}$$

Different λ histories generally give different luminosity distances. The cosmological source-evolution principle selects λ = 1 in the same reference T used by capacity. It removes this freedom and returns the preceding optical equation. The source law, retained central shell kernel and capacity relation supply the coefficients; the source principle supplies the evolution pace.

$$\frac{dV_D}{dT} = Q_D \quad \Longrightarrow \quad \lambda = 1 \quad \Longrightarrow \quad D_L(z) = (1+z) D(z) \tag{54}$$

The retained realization

The conclusion has an explicit domain: fixed baryonic material mass, homogeneous density, the isotropic central shell response, a flat homogeneous spatial reading, an advancing null accumulation front and conserved photons with the stated beam area. The front’s initial state and the reception source inputs complete the calculation. Within these premises, the distance history is fixed.

The radius equation and this optical equation are two routes through the same system. The first displays accumulation and dilution; the second displays the distance measured by the luminosity calculation. Neither route adjusts the law from one supernova group to another.

The common-energy calculation later in the paper asks a further, precise question: what energy and work must the same evolving state carry? Its lapse constraint is compatible with this history when A E_exp = E_b. That correspondence is derived below, with the source pace still explicit.

14. The observable and a worked calculation

The archived test compares standardized brightness between supernova groups at different redshifts. It does not compare the fading of one supernova at two dates. Each target group is referenced to the catalogue's nearby sample, using the same contrast weights and published uncertainties as the earlier calculation.

Define d_L = H_0D_L/c, with H_0 = H_b0/α_0. Let $\mathscr{L}$ be the target-minus-reference weighted contrast, with $\mathscr{L}[1] = 0$. The constant-clock history supplies the dimensionless baseline. With the catalogue's heliocentric redshift in the prefactor and Hubble-diagram redshift in the integral,

$$d_{L,0} = 2(1 + z_{\mathrm{HEL}})\left[1 - (1 + z_{\mathrm{HD}})^{-1/2}\right] \tag{55}$$

$$d_L = (1 + z_{\mathrm{HEL}}) \int_0^{z_{\mathrm{HD}}} \frac{\alpha(z')}{\alpha_0 (1 + z')^{3/2}}\, dz' \tag{56}$$

$$B_{\mathrm{obs}} = 10^{-0.4\mathcal{L}[m - 5\log_{10} d_{L,0}]}, \qquad B_{\mathrm{SET}} = 10^{-0.4\mathcal{L}[5\log_{10}(d_L/d_{L,0})]} \tag{57}$$

A common additive magnitude offset and a common distance normalization cancel because the contrast has zero total weight. Predictions are evaluated at each catalogue redshift before the contrast is formed; a group's mean redshift is only its plotting label.

Pantheon+: mean redshift 0.730852

For the archived group of 63 target objects and its 311-object nearby reference, the baseline is B_0 = 1. The radius-derived history gives a contrast magnitude of 0.3769637728. The measured contrast is 0.3862377659. Therefore

$$B_{\mathrm{SET}} = 10^{-0.4(0.3769637728)} = 0.7066664790 \tag{58}$$

$$B_{\mathrm{obs}} = 10^{-0.4(0.3862377659)} = 0.7006560778 \tag{59}$$

The observed one-sigma interval is 0.679033 to 0.722968. The central discrepancy falls from 42.72% for the unit baseline to 0.86% for this SET calculation. This worked example uses the same rule as every group in the following table.

15. The fifteen reproduced brightness contrasts

The independent radius calculation reproduces the complete archived set below. $B_0 = 1$ for every group. The error is $100(B_{SET}/B_{obs} - 1)$; quoted intervals are the archived marginal one-sigma observational intervals.

Catalogue	Mean z	Observed [1σ]	SET	Error
Pantheon+	0.158	0.9198 [0.9024, 0.9375]	0.9370	+1.87%
Pantheon+	0.273	0.8428 [0.8272, 0.8586]	0.8790	+4.30%
Pantheon+	0.413	0.7798 [0.7667, 0.7931]	0.8170	+4.78%
Pantheon+	0.567	0.7444 [0.7268, 0.7623]	0.7587	+1.93%
Pantheon+	0.731	0.7007 [0.6790, 0.7230]	0.7067	+0.86%
Pantheon+	0.902	0.6720 [0.6194, 0.7290]	0.6613	-1.58%
Pantheon+	1.122	0.4941 [0.4546, 0.5371]	0.6141	+24.27%
Pantheon+	1.530	0.5639 [0.5237, 0.6072]	0.5512	-2.25%
DES-Dovekie	0.161	0.9547 [0.9307, 0.9794]	0.9358	-1.98%
DES-Dovekie	0.286	0.8572 [0.8405, 0.8743]	0.8732	+1.87%
DES-Dovekie	0.422	0.7899 [0.7711, 0.8092]	0.8138	+3.02%
DES-Dovekie	0.573	0.7332 [0.7193, 0.7474]	0.7574	+3.30%
DES-Dovekie	0.706	0.6940 [0.6808, 0.7075]	0.7146	+2.96%
DES-Dovekie	0.872	0.6624 [0.6446, 0.6808]	0.6691	+1.02%
DES-Dovekie	1.064	0.6567 [0.6280, 0.6866]	0.6254	-4.76%

All fifteen central discrepancies are smaller than the unit-baseline discrepancies. The median absolute percentage discrepancy changes from 38.53% to 2.09% for Pantheon+, and from 36.39% to 2.96% for DES-Dovekie. Five of eight and three of seven predictions, respectively, lie within their marginal one-sigma intervals.

Independent reconstruction	Groups	Maximum |new − archived|
Pantheon+	8	7.06335×10^{-12}
DES-Dovekie	7	3.71803×10^{-12}

The new result is algebraic and numerical reconciliation: the explicit source-radius history gives the existing brightness calculation. The table reuses earlier observations and is not an additional independent empirical sample. Catalogue correlations and shared calibration information are not removed by reproducing the predictions.

16. The brightness trend in the same calculation

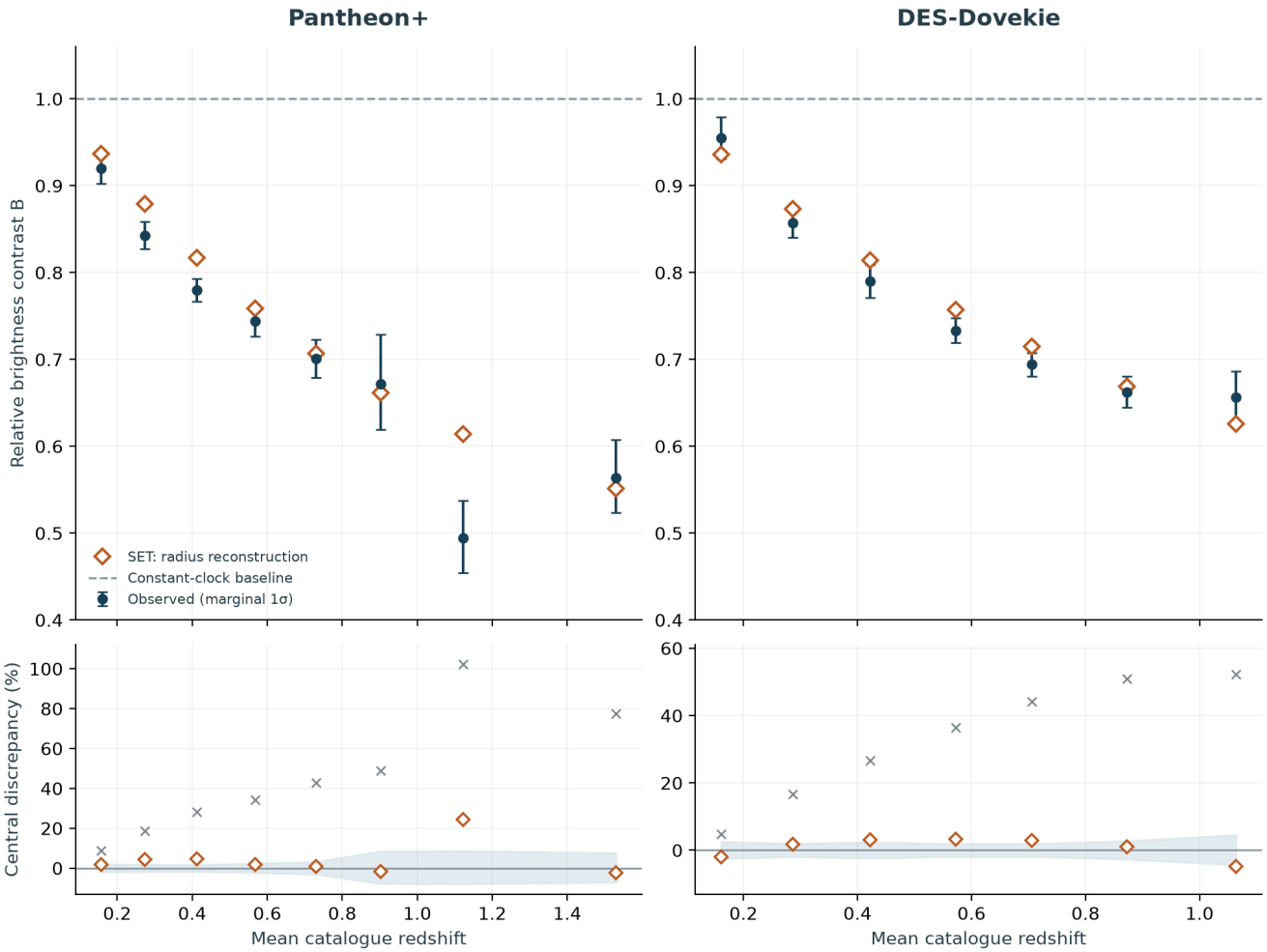


Figure 2. Archived grouped brightness contrasts and their reproduced SET predictions. Observations carry their published-analysis marginal one-sigma intervals. SET values are evaluated object by object before grouping. The horizontal line is the unit constant-clock baseline. Below, diamonds show SET discrepancies and gray crosses show baseline discrepancies. Shading shows the fractional observational one-sigma intervals about zero. This is a reproduction of the existing test, not a new fit or new sample.

The change is systematic across the two catalogue constructions: the source-clock history lowers the predicted relative brightness in the direction required by every central contrast. The table preserves the remaining differences, including the larger Pantheon+ residual near mean redshift 1.12.

The point of this derivation is that the brightness rule is now traceable through an explicit source radius, density, mass and clock history. The fifteen numerical predictions survive that change of mathematical route.

17. The initial source state and the front’s history

The exact integral also specifies the source-front history implied by the frozen reception state, when the zero-front branch is retained:

$$\frac{a_i}{a_0} = 0.2684965101, \qquad \frac{T_{Qi}}{T_{Q0}} = 0.1391258939 \tag{60}$$

$$T_{Qi} = 6.05953471 \text{ Gyr}, \qquad t_0 - t_i = 25.49724618 \text{ Gyr} \tag{61}$$

The first duration uses source-reference units; the second integrates the material clock dt = α dT_Q. They are model deductions from the chosen benchmark and history, not measured cosmic ages. The zero-radius front begins at finite representative-domain density and nonzero material scale factor.

Why the source start cannot be removed by a clock label

Define f(X) = −X/3 + √(1−2X²/3). On the positive-capacity interval,

$$f'(X) = -\frac{1}{3} - \frac{2X/3}{\sqrt{1 - 2X^2/3}} < 0 \tag{62}$$

The function crosses zero only at X_* = 3/√7. Below that point, a trajectory reaches X = 0 in finite backward ln T_Q. Above it, a trajectory reaches the capacity endpoint in finite backward ln T_Q. Hence only the fixed trajectory extends to every T_Q > 0 while keeping a nonnegative radius and a positive real clock.

$$u = \frac{6}{7}: \qquad t = \frac{T_Q}{\sqrt{7}}, \qquad R_h = 3ct, \qquad a \propto t^{2/3} \tag{63}$$

That self-similar trajectory has a constant clock factor and the constant-clock baseline distance shape. The varying branch used in the brightness calculation is a transient source-front history. Extending it to a formal zero-volume origin would require a different initial field state or a different physical regime; it is not achieved by shifting the clock’s origin.

Exact moving-volume bookkeeping

Integration from the retained launch to reception gives 54.7297% of V_h0 from the Q_normalized,h dt term and 45.2703% from cA_h dt. They sum to the full front volume. These are the two contributions in the moving-boundary identity, rather than two kinds of matter.

18. Common energy and the volume constraint

The source history can also be read through SET's common field energy. This connects the luminosity calculation to the same requirement used for material motion: clocks, motion and the source response belong to one accounting. Keep the conserved material domain explicit, writing V = V_D, E_b = M_Dc² and κ = 6πG/c². The volume kinetic form below is retained from the previously examined common-energy field class.

$$C = -\kappa V P_V^2 + E_{\mathrm{exp}} = 0, \qquad E_{\mathrm{exp}} = E_b + F \tag{64}$$

Here P_V is conjugate to volume. F denotes the actual retained motion, organization and spatial-response terms, with their field variables and momenta. In this retained reduction, F depends on volume and the remaining field variables and momenta, but not on P_V or the lapse. Its value is not replaced by the luminosity target before variation. E_exp names the positive expansion-side constraint energy; it is distinct from an isolated source's normalized boundary energy. The source law enters this same constraint through

$$Q_b^2 = 24\pi G M_D V = 4\kappa V E_b \tag{65}$$

Use an arbitrary evolution parameter s and the common positive lapse n, with dt = n ds and V_s = dV/ds. Eliminating only P_V from the retained canonical action gives its volume part:

$$L_V = -\frac{V_s^2}{4\kappa n V} - n E_{\mathrm{exp}} \tag{66}$$

The remaining canonical field terms keep their own equations. Vary n before fixing the reference clock. The lapse equation and its positive-volume branch give

$$\frac{V_s^2}{4\kappa n^2 V} = E_{\mathrm{exp}}, \qquad \frac{dV}{dt} = \sqrt{4\kappa V E_{\mathrm{exp}}} \tag{67}$$

This equation determines the material-clock volume pace of a supplied complete field state. It keeps the source response inside the energy balance rather than introducing a separate brightness correction. Changing the evolution parameter changes n and V_s together; their ratio remains the same physical volume rate.

An internal source phase must be transformed along with an energy-dependent source-volume clock. Keeping that canonical transformation returns dt/dT_Q = √(E_b/E_exp), in agreement with the volume equation. E_exp is therefore not substituted for the Hamiltonian conjugate to the internal phase.

19. What the lapse comparison selects

Keep the reference T general for one step. Write q = E_exp/E_b and λ = V_T/Q_b, where V_T = dV/dT. The ratio λ records the source pace in this reference; it is not an added coupling. The preceding constraint gives

$$n^2 q = \lambda^2, \qquad n \equiv \frac{dt}{dT} \tag{68}$$

There is an exact positive comparison associated with this lapse variation. At fixed volume rate and supplied field state, the volume action contains the cost nE_exp + λ²E_b/n. Subtract its stationary value and define

$$\mathcal{G}_\lambda = nE_{\mathrm{exp}} + \frac{\lambda^2 E_b}{n} - 2\lambda\sqrt{E_b E_{\mathrm{exp}}} \tag{69}$$

$$\mathcal{G}_\lambda = \frac{E_{\mathrm{exp}}}{n}\left(n - \lambda\sqrt{\frac{E_b}{E_{\mathrm{exp}}}}\right)^2 \geq 0 \tag{70}$$

Its unique zero on the positive branch is the lapse equation itself. This comparison comes from the retained volume action; it is not a new penalty chosen to have a desired minimum. With the capacity reading n² = A = 1−u, its zero is Aq = λ². Thus the lapse variation fixes the clock for the supplied source rate and energy.

The cosmological realization specifies the remaining source pace in the same reference: dV/dT = Q_b, so λ = 1 and T = T_Q up to its chosen origin. The shared capacity lapse is n = α. These retained source and clock premises therefore give

$$AE_{\mathrm{exp}} = E_b, \qquad H = \frac{Q_b}{3V\alpha}, \qquad F = \frac{E_b u}{1-u} \tag{71}$$

The final equality is the required on-history energy reading. It does not define F for arbitrary field states. This is the precise connection between the source-volume prescription, capacity and the common-energy constraint. The isolated spherical energy proof selects its exterior within different boundary conditions; the volume comparison here states its own variable and premises explicitly.

At the unchanged reception benchmark, E_exp/E_b = 4.16479720. This is the energy ratio required by the capacity clock and source pace, rather than a new fitted dark-energy density.

20. Source work and the complete-source reading

The selected history also specifies how the domain energy changes. Use ℓ = ln a, H = dℓ/dt and q = E_exp/E_b = 1/(1−u). Since E_b is conserved, the burden equation already derived in Section 7 gives

$$\frac{1}{q}\frac{dq}{d\ell} = g(q), \qquad g(q) \equiv \sqrt{6(q-1)} - (q-1) \tag{72}$$

In the retained domain decomposition, write X for the motion and organization contribution and Y for the spatial-response contribution. Their actual field exchange gives

$$E_{\mathrm{exp}} = E_b + X + Y, \qquad \dot{E}_{\mathrm{exp}} = H(Y - 3X) \tag{73}$$

A dot uses the material clock t. The luminosity history consequently fixes both required energy fractions:

$$\frac{X}{E_{\mathrm{exp}}} = \frac{1 - q^{-1} - g(q)}{4}, \qquad \frac{Y}{E_{\mathrm{exp}}} = \frac{3(1 - q^{-1}) + g(q)}{4} \tag{74}$$

These are requirements on the retained field terms, not additional kinds of matter. They make the proposed collective response testable in the common energy: the same state must supply its magnitude and its changing work. The relation involving Y−3X is therefore downstream of the history already selected by the source, capacity and front prescription.

The complete Q reading

$$Q_{\mathrm{exp}}^2 = \frac{24\pi G}{c^2} V E_{\mathrm{exp}}, \qquad Q_{\mathrm{exp}} = \frac{dV}{dt} = \frac{Q_b}{\alpha} \tag{75}$$

In the flat homogeneous spherical reading, E_exp also equals the spherical complete-energy reading $c^2R^3H^2/(2G)$, with $V = 4\pi R^3/3$. The equality does not identify it with an isolated boundary Hamiltonian. Its Q_exp is already the material-clock volume rate; applying another division by α would count the same clock conversion twice.

This completes the energy connection of the retained luminosity calculation. A joint realization in which one local field state supplies these X and Y histories, its capacity reading and the assumed photon and beam propagation has not been shown here. The equations above state exactly what that realization must reproduce while preserving the derived luminosity history.

21. What this derivation establishes

The collective-source picture now has a connected mathematical route to received brightness. The finite-domain source law gives Q. The cosmological source-evolution principle assigns the material volume pace in the common reference time. The uniform central response supplies S_c² = 3GM_h/R_h. The accumulating null front and material dilution determine its changing burden. Capacity supplies the clock progression used in photon propagation.

$$Q^2 = 24\pi GMV, \quad V_T = Q, \quad \alpha^2 = 1 - u \quad \longrightarrow \quad R_h,\ u,\ D_L \tag{76}$$

The resulting history can be followed through the source radius or directly through the optical capacity equation. Its source-photon identity makes the redshift leg explicit. The same luminosity rule reproduces all fifteen archived contrasts with the source inputs and calculation protocol held fixed. The new derivation changes the mathematical route, not the observational sample.

The energy connection

In the retained common-energy volume class, lapse variation gives A E_exp/E_b = λ². The cosmological source principle fixes λ = 1 and therefore selects A E_exp = E_b. The exact nonnegative lapse comparison identifies this equality as the stationary result of that retained constraint. The work fractions then state the energy evolution required along the selected history.

$$AE_{\rm exp} = E_b, \qquad \frac{\dot{E}_{\rm exp}}{HE_{\rm exp}} = \sqrt{6(q-1)} - (q-1), \quad q = \frac{E_{\rm exp}}{E_b} \tag{77}$$

The gain is a stated physical premise followed by a reproducible chain of consequences. The selection is within the cosmological realization specified here. It does not establish that the scalar capacity invariant alone excludes every field structure. Appendix B locates the compatibility condition for a local interacting realization, keeping its actual field evolution distinct from the energy ratio required by the luminosity solution.

The source law, capacity clock and source-evolution principle together make this SET cosmology concrete: ordinary baryonic sourcing produces a changing collective response, that response enters propagation and arrival, and those quantities give the measured brightness comparison.

Appendix A. Source production and a moving volume

Earlier SET described Q as emitted space volume per reference time and stated that total cosmological flux supplies volume growth. That is the historical antecedent of the source pace retained here. Applying it to the material volume is a physical identification; the volume emitted outside a stationary body is not the volume occupied by that body.

The earlier four-dimensional flux equation provides a useful bookkeeping check. In its flat coordinates, write $x^0 = cT$ and $F^0 = cb$, with dimensionless b and spatial flow w in speed units. Its full divergence equation reads

$$\partial_T b + \nabla \cdot \mathbf{w} = \chi, \qquad \chi \equiv \sqrt{24\pi G\rho} \tag{78}$$

Let Ω move with material-boundary velocity v_m. Define its integrated production, temporal-storage contribution and boundary-slip contribution by

$$Q_{\text{int}} = \int_\Omega \chi \, dV, \quad P_{\text{time}} = \int_\Omega \partial_T b \, dV, \quad J_{\text{slip}} = \oint_{\partial\Omega} (\mathbf{v}_m - \mathbf{w}) \cdot \mathbf{n} \, dA \tag{79}$$

All three have volume-per-time units. The divergence theorem and the moving-boundary volume identity then give

$$\frac{dV}{dT} = Q_{\text{int}} - P_{\text{time}} + J_{\text{slip}} \tag{80}$$

For uniform density, $Q_{int} = \sqrt{(24\pi G\rho)}V = Q_D$. Hence the source pace has the exact diagnostic form

$$\lambda \equiv \frac{V_T}{Q_D} = 1 + \frac{J_{\text{slip}} - P_{\text{time}}}{Q_D}, \qquad \lambda = 1 \iff J_{\text{slip}} = P_{\text{time}} \tag{81}$$

This identity distinguishes source production from material-volume change. The spatial-only equation follows when the temporal term vanishes. It cannot be used for a changing field by silently dropping that term. Likewise, calling matter comoving does not equate its boundary velocity with an independently specified flow.

The calculation above does not identify b with α or reinstate the earlier four-vector dynamics. It shows why the modern source-volume pace must state its domain and clock explicitly. Retaining that pace selects the luminosity prescription developed in the main text; this bookkeeping identity is not a separate derivation of it.

Historical source: E. Echeverria, Space Emanation Theory, Rough Draft 22, “Flux-mass continuity” (pp. 13-14), “Flux action and field equations” (pp. 28-30), and the cosmological volume-growth discussion (p. 134).

Appendix B. Regular launch and field compatibility

The source-paced history gives a precise requirement for its realization in the retained tensor dynamics. Consider a fixed material domain with constant baryonic energy E_b, N = ln a and H = dN/dt > 0. In the homogeneous canonical class, let K be its smooth nonnegative quadratic shape kinetic term and U its spatial-response term:

$$X = \frac{K}{V}, \qquad Y = V^{1/3}U, \qquad E_{\mathrm{exp}} = E_b + X + Y, \qquad q = \frac{E_{\mathrm{exp}}}{E_b} \tag{82}$$

At a regular finite-density front launch, R_h = 0 and dR_h/dt = c. The common constraint at q_i = 1 gives H_i² = 8πGρ_i/3. Together with the central kernel, this yields u = (3/2)(N−N_i)² to leading order. With the source-paced relation q = 1/(1−u), this requires

$$q_i = 1, \qquad q_{N,i} = 0, \qquad q_{NN,i} = 3 \tag{83}$$

Canonical evolution, retaining the exchange J = {K,U} between both field terms, gives

$$q_N = \frac{Y - 3X}{E_b}, \qquad q_{NN} = \frac{Y + 9X}{E_b} - \frac{4J}{HV^{2/3}E_b} \tag{84}$$

The first two launch conditions force X_i = Y_i = 0. Since K is smooth and nonnegative in this class, K_i = 0 is a minimum: dK_i = 0 and therefore J_i = 0. The actual field evolution then gives q_NN,i = 0. Its exact energy and work balance also yields

$$\frac{d}{dN}\left[e^{-N}(q-1)\right] = -\frac{4e^{-N}X}{E_b} \le 0 \tag{85}$$

Thus this homogeneous positive-kinetic class cannot generate the rising source-paced history from its regular launch. The field realization must carry a response beyond that restricted class; assigning the required energy ratio alone does not supply its evolution.

The full inhomogeneous field permits signed effective kinetic response from spatial variations in expansion, so the nonnegative-K argument does not exclude that larger class. The luminosity calculation remains selected by the stated source pace, capacity and front prescriptions. This appendix identifies the compatibility requirement for extending that selection to an actual tensor-field source.

Sources, checks and the preserved calculation

Reproducibility

Rough Draft 1 independently integrated the source radius, reconstructed the front mass and evaluated the null distance separately from the archived y(z) brightness code. The reception inputs, catalogue redshifts, reference groups and contrast weights were held fixed. The maximum differences from the archived predictions were 7.06335×10^{-12} for Pantheon+ and 3.71803×10^{-12} for DES-Dovekie. Rough Draft 2 preserves both tables and both figures from that manuscript.

The new results are algebraic: the explicit source pace, the source-photon identity, the direct optical equation, the general λ diagnostic and the lapse-energy comparison. The appendices retain temporal storage in the historical transport equation and test the regular launch in a specified homogeneous kinetic class. These checks establish mathematical connections and their scope; they do not create additional empirical observations.

SET sources

Echeverria, E. Space Emanation Theory - Source, Capacity and the Luminosity History, Rough Draft 1 (23 October 2026). Source-radius derivation, reception benchmark and the archived fifteen-contrast reconstruction.

Echeverria, E. Space Emanation Theory - Integrated Rough Draft 5. Source and capacity framework, homogeneous luminosity application and the catalogue protocol. The numerical corpus reproduced here was already present in Integrated Rough Draft 4.

Echeverria, E. Space Emanation Theory - Source, Energy, Effective Area and Tidal Response, Rough Draft 1. Companion source-domain and energy-accounting analysis.

Echeverria, E. Space Emanation Theory, Rough Draft 22. Historical volume-production language; “Flux-mass continuity,” pp. 13-14; “Flux action and field equations,” pp. 28-30; cosmological volume growth, p. 134. The earlier formulation is used for provenance and the explicit bookkeeping test in Appendix A.

Research record

Research archive, 23-24 October 2026: “From the density source law to the luminosity history”; the luminosity-selection premise audit; optical and common-energy inevitability checks. The observational benchmark remains verified-results.json from Space Emanation Theory - Dark Energy, Rough Draft 3. The complete archived set is displayed in this paper, including the larger residuals.